STANDATA interpretation 23-BCI-025/24-ECI-009

Building/Electrical

Service/utility room for energy storage systems

Date Issued: October 2026

Purpose

To clarify that a building safety codes officer (SCO) reviews and inspects the design and construction of a room containing energy storage systems (ESS) where the room provisions are in the 2024 Canadian Electrical Code, Part 1 (CE Code Part 1).

Discussion

Electrical and building SCOs have raised concerns over which discipline administers the requirements in CE Code Part 1, Rule 64-1100 1) d), 2) and 11), pertaining to the construction of a room containing an ESS.

The *Safety Codes Act* (SCA) along with its 21 regulations establishes a unifying framework for the administration of 11 safety disciplines, with each having their own safety codes and standards.

An SCO administers the SCA, the regulations reference the applicable codes which fall within the scope of administering the SCA.

Under Clause 6(1) of the Permit Regulation, a permit in the building discipline is required for the construction of a building, including the renovation of or addition to a building. As such, if construction in the building discipline is undertaken, a building permit is required.

The National Building Code - 2023 Alberta Edition (NBC(AE)) references CAN/CSA-B72, “Installation Code for Lightning Protection Systems.” A lightening protection system is an electrical system falling within the scope of the Electrical SCO for permit and inspection.

Code References

C22.1:24- 2024 Canadian Electrical Code, Part 1

Section 0 — Object, scope, and definitions

Definitions

Service room — a room or space provided in a building to accommodate building service equipment and constructed in accordance with the National Building Code of Canada or applicable local legislation (see Appendix B, Note to Rule 26-010).

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Section 64 — Renewable energy systems, energy production systems, energy storage systems, and batteries

Installation of energy storage systems at residential occupancies

64-1100 Location and separation requirements (see Appendices B and G)

1) Except as required by Subrule 2), energy storage systems installed at a dwelling unit or building of residential occupancy shall be suitable for residential use, and be located
   a) in an attached garage;
   b) in or on an associated detached garage, or other freestanding structure;
   c) on the exterior surface of the building;
   d) in a dedicated room or utility room having a door equipped with a self-closing device and enclosed with a minimum construction of
      i) ceilings and walls finished with gypsum board; and
      ii) floors finished with lumber sheathing; or
   e) in other locations where permitted.
2) Where an energy storage system is installed in a building of residential occupancy containing multiple dwelling units, the construction required in Subrule 1) d) shall have a fire resistance rating of not less than 1 h.

11) Where an energy storage system is installed in accordance with Subrule 1) d), a smoke alarm or smoke detector shall be installed;
   a) in the room where the energy storage system is located; and
   b) in accordance with Section 32.

Appendix B
Rule 64-1100 1)
The fire resistance rating of ceilings, floors, and walls constructed for the purposes of this Rule should be verified by a qualified individual in accordance with the requirements of the authority having jurisdiction prior to installation.

Additional information regarding the standard methods of fire endurance tests of building construction and materials can be found in CAN/ULC-S101. This Standard covers fire endurance tests applicable to walls, partitions, floors, roofs, ceilings, columns, beams, and girders, as well as some components of these building sub-assemblies.

Additional information regarding fire resistance values of generic wall and floor assemblies can be found in Table 9.10.3.1.A and 9.10.3.1.B of the *National Building Code of Canada* or in Appendix D of the *National Building Code of Canada*.

An example of a wall providing a 1 h fire resistance rating is a construction of 38 mm × 89 mm wood studs spaced 400 mm or 600 mm o.c. without absorptive material in the cavity and with one layer of 15.9 mm Type X gypsum board on each side.

An example of a floor/ceiling assembly providing a 1 h fire resistance rating is a construction of a typical wooden subfloor [15.5 mm plywood, Oriented Strand Board (OSB) or waferboard, or 17 mm tongue and groove lumber] on wood joists or wood I-joists spaced not more than 400 mm o.c. and two layers of 12.7 mm Type X gypsum board on the ceiling side without absorptive material in cavity. Where the spacing of the joists is 600 mm, two layers of 15.9 mm Type X gypsum board would be required to achieve the same rating.

Alternative locations for wall, ceilings and floor assemblies may be considered, subject to acceptance from the local building, fire, or other appropriate authority having jurisdiction.

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Appendix G

CE Code Sections and Rules	NBCC Sections, Articles, and Sentences (unless otherwise stated, references refer to Division B of the NBCC)
Rule 64-1100	9.10.3.1., Fire resistance and fire protection ratings
Rule 64-1100	Appendix D, Fire performance ratings

Safety Codes Act

Owners, care and control

5 The owner of any thing, process or activity to which this Act applies shall ensure that it meets the requirements of this Act, that the thing is maintained as required by the regulations and that when the process or activity is undertaken it is done in a safe manner

Contractors' duties

8 A contractor who undertakes construction, operation or maintenance of or builds or installs any thing to which this Act applies shall ensure that this Act is complied with.

Permit Regulation pursuant to the Safety Codes Act

Interpretation

1(1) In this Regulation,

(v) "undertaking" means the construction of a thing or the control or operation of a thing, process or activity to which the Act applies.

Permit required

3(1) Subject to subsection (2), a person shall not start any undertaking for which a permit is required under this Regulation unless a permit has been issued.

Building permit

6(1) A permit in the building discipline is required for the following:
(a) the construction of a building, including the renovation of or addition to a building;
(b) a change in the occupancy classification of a building.

Electrical permit

8(1) A permit in the electrical discipline is required to install, alter or add to an electrical system.

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Application

This interpretation applies to the review and inspection of a service/utility room for the purpose of complying with CE Code Rule 64-1100 1)d), 2) and 11)

Interpretation

1. A room is required by the CE Code to enclose an ESS; the construction of the room is to be reviewed and inspected for compliance with the CE Code by a building SCO.

2. In addition to an electrical permit being required for the electrical installation of an ESS, a building permit is required for the construction of room containing an ESS.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to the 2024 Canadian Electrical Code, Part 1 and the National Building Code - 2023 Alberta Edition (NBC(AE))

Issued by the Provincial Electrical and Building Administrators

[Original signed]
Kevin Glubrecht
Provincial Electrical Administrator

[Original signed]
Paul Chang
Provincial Building Administrator

Alberta Municipal Affairs – Technical and Corporate Services
Email: safety.services@gov.ab.ca Phone: 1-866-421-6929

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